Annex Extra — What is missing
Why so many small positive studies have not yet translated into clinical recommendations?

An in-depth discussion on the translational gap in hydrogen medicine — from signal to standard of care

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1. The paradox: many signals, no recommendation

In 18 years, molecular hydrogen accumulated the classic translational promise profile: thousands of publications, 81 registered trials, consistent safety in 34 RCTs (no serious AEs attributed) and dozens of statistically positive results — total cholesterol −6.7 mg/dL, LDL −3.2 mg/dL, BAP SMD 0.29, explosive power SMD 0.30, RPE −0.37. Yet the conclusion of this review — and of Johnsen 2023, Dhillon 2024 — is the same: no basis for clinical recommendation. The only adequately powered phase 3 RCT (Hydro-COVID, n=675, triple-blind) was negative. This mismatch is not fraud: it is the expected behavior of a field stuck at early signals, before definitive testing.

Core point: statistical significance ($p<0.05$) is not clinical significance, and signal on a surrogate is not benefit on a hard outcome. H2 medicine is repeating the story of antioxidants, vitamin D and omega-3: many positive pilots, few confirmatory giants.

2. Seven barriers blocking translation

Why do so many $p<0.05$ not become guidelines? We group seven interdependent barriers — all present in the H2 evidence base:

Barrier	What it is	How it appears in H2	Concrete example	What would unlock
1. Small sample & winner's curse	RCTs n=12–60 overestimate effect when positive	Median n=28 in 34 RCTs; pilots n=12–14 published as positive	Korovljev 2019 (n=12, NAFLD) → 18% liver fat ↓, no replication	RCTs ≥200 & PROSPERO
2. Non-standard dose	No dose-response, no PK	HRW 0.5–5.9 ppm vs gas 1–66%; 9 RCTs no ppm	Performance: same HRW 0.5 vs 5.9 ppm pooled as equal	Phase 1 PK + head-to-head dose-finding
3. Surrogate endpoint	BAP, d-ROMs, LDL −3% ≠ clinical event	BAP SMD 0.29 p=0.03 but d-ROMs 0; LDL −6.7 mg/dL (<5% vs 30–50% statin)	LDL −3.2 mg/dL statistically positive, clinically irrelevant	Core Outcome Set with biopsy, MACE, function
4. Publication bias & short follow-up	Positive pilots published, negatives filed; short	4–24 weeks; max 24 w (Zanini); no ≥12 mo; Egger not assessable	Hydro-COVID (only phase 3, n=675) negative — only powered to refute	Mandatory registration + funnel ≥10 + 12 mo
5. Weak blinding/randomization	Inhalation unblinded; randomization not described	RoB2: 15/34 no details; 7 inhalation unblinded	Ito 2024 (2% inhal.): participants unblinded	Sham air + blinded assessor + central allocation
6. Homogeneous population	>90% E. Europe/Asia	Serbia, Japan, Slovenia dominant	Zanini 2021 (70+ Italians) not replicated	Multicenter with ethnicity/diet/microbiota stratification
7. Inverted incentive	Non-patentable → little pharma for costly phase 3	Commercial products before evidence; MHI-linked studies	LeBaron 2020 (MHI) 5.9 ppm — sensitivity excludes	Public funding (NIH/Horizon) for independent phase 3

Note: GRADE downgraded cholesterol/LDL to Moderate for imprecision (CI crosses MCID 10 mg/dL and n<400) and RPE to Very low for I^2 58% + imprecision. Hydro-COVID kept High (negative) as only OIS-reached (n=675, 80% power for HR 0.6).

3. The small-effect and surrogate problem

In GRADE, effect is not enough: it must exceed MCID and come from a patient-important outcome. This is where H2 stumbles. Our Box 2 shows total cholesterol −6.7 mg/dL and LDL −3.2 mg/dL — $p<0.05$ and $p=0.043$, but MCID is ~10 mg/dL; statin reduces 30–50%. Patients do not feel −3 mg/dL, and no guideline changes for it. For performance, BAP SMD 0.29 without d-ROMs fall suggests potentiation of endogenous defense, mechanistically interesting, but BAP/d-ROMs are lab, not VO2max or injury. Explosive power SMD 0.30 and RPE −0.37 are small (Cohen) and heterogeneous (I^2 58%). In NAFLD, pilots with 12–18% relative liver fat reduction on MRI look attractive, but without biopsy, multicenter replication and n=12–60, GRADE falls to Low. Common error is confusing biological signal with clinical benefit. H2 has the former; lacks the latter.

MCID example: for cholesterol we used 10 mg/dL (NCEP-ATP III) and OIS n=400. LDL −3.2 mg/dL [95%CI −6.31 to −0.10] crosses threshold and n=757 but 56% RCTs have some concerns — so GRADE downgrades to Moderate. For SMD, MCID=0.30 (Cohen small-moderate); BAP 0.29 is at threshold, RPE −0.37 with I^2 58% falls to Very low.

4. Why H2 suffers more than other nutraceuticals?

Three factors make H2 especially vulnerable to the translational valley of death. First, it is non-patentable and cheap — great for access, terrible for attracting pharma investment of hundreds of millions for a 12-month hard-outcome phase 3 (MACE, biopsy). Second, route/dose is chaos: pooling HRW 0.5 ppm ad libitum with 66% inhalation 60 min as "H2" is like pooling 1 mg and 100 mg of a drug as "drug". Without standardized human PK (half-life 30–60 min, exhaled H2 as biomarker, adjustment for basal microbiota), any meta-analysis mixes apples and oranges — hence I^2 58% for RPE. Third, the field grew via supplement/commercial before academia: 81 registered RCTs but median n=28 and 9 RCTs not reporting concentration. Commercial incentive pushes positive pilots; the large negative (Hydro-COVID) is the exception. Same script as vitamin D and omega-3 in the 2010s: hundreds of enthusiastic observationals/pilots, then VITAL and REDUCE-IT showed true effect size.

5. What is missing to recommend? Translational checklist

In GRADE and guideline criteria (e.g., ACC/AHA, AASLD), recommendation requires 5 conditions. H2 today meets only 1:

Condition to recommend	H2 today	What is needed
1. Safety in large diverse sample	☑ **Met (34 RCTs, 0 serious)**	Keep active pharmacovigilance
2. Benefit on hard or validated surrogate	✗ **No (LDL −3%, BAP, RPE — none validated)**	MACE, NASH biopsy, fracture, hospitalization — with MCID
3. Clinically relevant & precise effect	✗ **No (CI crosses MCID, n<400)**	DM ≥10 mg/dL or SMD ≥0.5 with narrow CI
4. Optimal dose/route & PK defined	✗ **No (0.5–5.9 ppm vs 1–66% no curve)**	Phase 1 PK + head-to-head dose-finding HRW vs gas
5. Independent multicenter replication	✗ **No (pilots n=12–60, single-center, >90% Asia/E. Europe)**	≥2 phase 3 RCTs, n>200, ≥12 mo, multicenter

6. Roadmap out of the valley of death

Phase 0–1 (6–12 mo) — Pharmacokinetics — Crossover n=20–30 healthy: HRW 0.5/1.5/5.9 ppm vs 2%/66% inhalation vs HRS; exhaled/blood H2 every 10 min for 2h, BAP/d-ROMs, adjust for microbiota (lactulose test) and diet. Define Cmax, Tmax, AUC and adherence biomarker.

Phase 2 (12–18 mo) — Dose-finding — 4-arm RCT n=120 (30/arm) in metabolic syndrome: placebo vs low/mid/high HRW vs 2% gas; outcome: LDL + BAP with pre-defined MCID; select 1–2 doses for phase 3.

Phase 3 (24–36 mo) — Confirmatory efficacy — Two parallel RCTs n=250 each: (A) NAFLD with biopsy (NAS) at 12 mo; (B) performance/fatigue with VO2max + time to exhaustion at 6 mo; both ≥12 mo, multicenter, PROSPERO, core outcome set, ITT, result-independent publication.

Follow-up — Registration & transparency — Prospective PROSPERO, published SAP, individual data in Zenodo/OSF, sensitivity by funding (MHI) and by reported concentration.

Proposed timeline: PK (0–1 y) → Dose-finding (1–2.5 y) → Phase 3 (2.5–5.5 y) → Guideline only if ≥2 positive replicated phase 3

7. Conclusion: from enthusiasm to discipline

Hydrogen medicine does not need more positive pilots with n=20 and p=0.03; it needs fewer, larger, better, preregistered studies. The field has proven safety and generated an elegant mechanistic hypothesis (Fe-porphyrin → Keap1 → Nrf2). Now it must prove — or refute — patient-important benefit, at the right dose, via the right route, long enough. Meanwhile, the honest recommendation is as in the manuscript: do not recommend routinely, do not delay proven therapy and, if the patient already uses it, inform cost and uncertainty transparently. The next chapter will not be written with more enthusiasm, but with methodological discipline. This annex is the invitation to that chapter.

Message for clinicians: today, H2 is a research question, not a therapeutic answer. For researchers: the question deserves a well-designed 5-year program, not 5 more years of pilots.

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